

Dr. Katherine R. Hayes
krhayes.com | hayesk7@wwu.edu
ORCID: 0000-0002-6612-9398

RESEARCH APPOINTMENTS

- Present** Assistant Professor of Forest Systems
Dept. of Environmental Science, Western Washington University
- 2025 - 2026** Term Assistant Professor of Forest Ecology
Dept. of Natural Resources, University of Alaska Fairbanks
- 2023 - 2025** National Science Foundation Postdoctoral Research Fellow
Forest Futures Lab, Cary Institute of Ecosystems Studies

EDUCATION

- 2018 - 2022** Ph.D. in Integrative and Systems Biology
University of Colorado, Denver
Advised by Dr. Brian Buma
- 2016 - 2018** M.S. in Geography
University of Oregon
Advised by Dr. Dan Gavin
- 2012 - 2016** B.S. in Environmental Studies, Spanish, and Geography, with honors
University of Wisconsin, Madison
Advised by Dr. Jack Williams

PEER-REVIEWED PUBLICATIONS

14. Conway A, Hayes K, Jean M, Ruess R, Johnstone J. Moderate responses of early tree growth to herbivore exclusion following fire in Alaskan boreal forests. *Canadian Journal of Forest Research*. [In Press]
13. Weiss S, Rancher W, Hayes K, Buma B, Lucash M. Climate change may lead to larger fires, greater area burned, and more reburning across interior Alaska. *Canadian Journal of Forest Research*. [In Press]
12. Barrio, I, Vuorinen K, Barbero Palacios L, Defourneaux M, Petit Bon M,... Hayes K, ... Kamena S. Emerging priorities in terrestrial herbivory research in the Arctic. (2025). *Arctic Science*. <https://doi.org/10.1139/as-2024-0080>
11. Carter T, Stears A, Fornwalt P, Atkins D, Dwire K, Fleri J, Hayes K, Mount H, Twaddell E, Wessel S, Buma B, Laughlin D. Fine-root traits coordinate with aboveground strategies yet poorly predict species' response to spruce mortality. (2025). *Plant Soil*. <https://doi.org/10.1007/s11104-025-07463-x>
10. **Hayes K**, Hoffman C, Linn R, Ziegler J, Buma B. Fuel constraints, not fire weather conditions, limit fire behavior in reburned boreal forests. (2024). *Agricultural and Forest Meteorology*. <https://doi.org/10.1016/j.agrformet.2024.110216>
9. Weiss S, Marshall A, **Hayes K**, Nicolsky D, Buma B, Lucash M. Future transitions from a conifer to a deciduous-dominated landscape are accelerated by greater wildfire activity and climate change in interior Alaska. (2023). *Landscape Ecology*. <https://doi.org/10.1007/s10980-023-01733-8>
8. Carter T, **Hayes K**, Buma B. Putting more fuel on the fire... or maybe not? A synthesis of spruce beetle and fire interactions in North American Subalpine Forests. (2022). *Landscape Ecology*. <https://doi.org/10.1007/s10980-022-01481-1>

7. **Hayes K**, Carter T, Cook P, Twaddell E, Buma B. Supporting Graduate Field Leadership through Community-Sourced Advice, Action, and Policy. (2022). *Ecosphere*. <https://doi.org/10.1002/ecs2.4247>
6. Buma B, **Hayes K**, Weiss S, Lucash M. Short interval fires increasing in the Alaskan boreal forest as fire self-regulation decays across forest types. (2022). *Scientific Reports*. <https://doi.org/10.1038/s41598-022-08912-8>
5. Muthukrishnan R, **Hayes K**, Bartowitz K, Cattau M, Harvey B, Lin Y, Lunch C. Harnessing NEON to evaluate ecological tipping points: opportunities, challenges, and approaches. (2022). *Ecosphere*. <http://doi.org/10.1002/ecs2.3989>
4. Nagy C, Balch J, Bissell E, Cattau M, Glenn N, Halpern B, ... **Hayes K**, ... O’Riordan, C. Harnessing the NEON data revolution to advance open environmental science with a diverse and data-capable community. (2021). *Ecosphere*. <https://doi.org/10.1002/ecs2.3833>
3. **Hayes K**, Buma B. Effects of short-interval disturbances continue to accumulate, overwhelming variability in local resilience. (2021). *Ecosphere*. <https://doi.org/10.1002/ecs2.3379>
2. Jensen A, Fastovich D, Watson B, Gill J, Jackson S, Russell J, Bevington J, **Hayes K**, Williams J. More than one way to kill a spruce forest: The role of fire and climate in the late-glacial termination of spruce woodlands across the southern Great Lakes. (2021). *Journal of Ecology*. <https://doi.org/10.1111/1365-2745.13517>
1. Buma B, Weiss S, **Hayes K**, Lucash M. Wildland fire reburning trends across the US West suggest only short-term negative feedback and differing climatic effects. (2020). *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/ab6c70>

OTHER PUBLICATIONS

- Reichenborn M, Rottler C, Aragón L, Bishop T, **Hayes K**, Kottler E, Talal M, Sommers P. Advice from the field: practical skills, challenges, and how to support early career ecologists. (2023). *Bulletin of the Ecological Society of America*. <https://doi.org/10.1002/bes2.2127>
- Kulakowski D, Buma B, Guz J, **Hayes K**. The ecology of forest disturbances. (2019). Reference Module in *Earth Systems and Environmental Science*. <https://doi.org/10.1016/B978-0-12-409548-9.11878-0>

PUBLICATIONS IN REVIEW (available on request)

- **Hayes K**, Lucash M, Olson K, Buma B. Repeat short-interval fires put carbon storage at risk in Interior Alaska via cumulative combustion of soil carbon. *Global Change Biology*. **[In Revision]**
- Carter, T*, **Hayes K*** **[Joint co-authors]**, Buma B. Boreal forest understory plant communities transform after one fire but demonstrate resilience to continued reburning. *American Journal of Botany*. **[In Revision]**
- Forsythe J, Mack M, Berner L, Goetz S, Hansen W, **Hayes K**, Johnstone J, Potter S, Rogers B, Talucci A, Wang J, Walker X. Short-interval wildfires are smaller, less severe, and require more extreme weather than long-interval fires in Boreal North America. *Fire Ecology*. **[In review]**

PRESENTATIONS

Invited Public Seminars and Webinars

- *Down the rabbit hole: modeling Alaskan herbivores and their interactions with fire*. iLand North America Working Group. Webinar. October 2025.
- *Understanding the mechanisms, impact and management of disturbances in boreal forests*. University of Alaska Fairbanks, Dept. of Natural Resources and Environment. Fairbanks, AK. Seminar. February 2025.
- *How herbivores interact with fire to shape Alaskan forests: challenges and opportunities in understanding disturbance interactions under climate change*. Cary Institute of Ecosystem Studies. Millbrook, NY. [Seminar](#). November 2023.

- *Frequent fires change forest composition and carbon of Alaskan boreal forests*. Natural Resources Institute Finland. Seminar. Helsinki, Finland. April 2023.
- *Evaluating flammability of reburns across Interior Alaska*. Alaska Fire Science Consortium. [Webinar](#). Nov. 2022.
- *How do we practice, develop and support graduate field leadership skills?* Pal(a)eo Early Career Seminar Series. [Webinar](#). September 2022.
- *Evaluating flammability of reburns in the boreal forests of Interior Alaska*. Alaska Fire Science Consortium. [Webinar](#). April 2020.

Oral presentations (presenting author only)

- 2024** Bonanza Creek LTER Science Symposium. Hayes K, Hansen W, Johnstone J. *Modeling fire-herbivore interactions in Interior Alaska*. Fairbanks, AK. June 2024.
- 2023** American Geophysical Union. Hayes K, Hansen W, Johnstone J, Honkaniemi J, Albrich K. *How do Interactions between Fire and Biotic Disturbances influence Forest Carbon and Communities in Interior Alaska?* San Francisco, CA. December 2023.
- Ecological Society of America. Hayes K, Ziegler J, Hoffman C, Buma B. *Simulated fire spread in reburned boreal deciduous forests limited by wind speed and fuel connectivity, not fuel availability*. Portland, OR. Aug. 2023.
- International Association of Landscape Ecology: North America. Hayes K, Hansen W. *A systematic review of emerging biotic disturbances in Interior Alaska*. Riverside, CA. March 2023.
- American Meteorological Society. Hayes K, Buma B, Ziegler J, Hoffman C. *Changing fuels and fuel structures in boreal forests alter and can increase fire spread through wind, spread, and other mechanisms*. Denver, CO. January 2023. **[Invited]**
- 2022** American Geophysical Union. Hayes K, Hoffman C, Ziegler J, Lucash M, Buma B. *Continued short-interval reburning changes carbon storage and future fire behavior of boreal forests regardless of forest resilience*. Chicago, IL. December 2022.
- Ecological Society of America. Hayes K, Lucash M, Hoffman C, Ziegler J, Buma B. *Short-interval reburning changes fuel structure, carbon storage and fire behavior of boreal forests*. Montreal, CA. August 2022.
 - Ecological Society of America: Inspire Session. Hayes K, Carter T, Cook P, Twaddell E, Buma B. *Strategies for managing and leading fieldwork successfully as a graduate student*. Montreal, CA. August 2022.
 - International Association of Landscape Ecology: North America. Hayes K, Lucash M, Buma B. *Repeat short-interval fires put carbon storage in Interior Alaska at risk*. Remote. April 2022. **[Presentation Award]**
 - Northwest Science Association. Hayes K, Gavin D. *Establishing baseline patterns of fire in old-growth coast redwood forests using soil carbon and charcoal*. Remote. March 2022.
 - Front Range Student Ecology Symposium. Hayes K, Carter T, Cook P, Twaddell E, Buma B. *Graduate field leadership: Challenges, successes and strategies*. Fort Collins, CO. Feb. 2022.
- 2021** American Geophysical Union. Hayes K, Buma B. *Repeat short-interval fires drive changes in biomass and soil carbon in Interior Alaska, regardless of local site conditions or resilience*. New Orleans, LA. Dec. 2021.
- Great Plains Rocky Mountains Applied Geography. Hayes K, Buma B. *Continued short-interval reburning drives changes in fuel-scapes across boreal forest landscapes*. Remote. October 2021.
 - International Boreal Forest Research Association. Hayes K, Buma B. *Biomass and soil carbon in reburned stands burnt in short-intervals in Interior Alaska*. Remote. August 2021. **[Presentation Award]**

- International Association of Landscape Ecology: North America. Hayes K, Buma B. *Repeat short-interval fires drive changes in biomass and soil carbon in Interior Alaska.* Remote. April 2021.
 - American Association of Geographers. Hayes K, Buma B. *The effects of multiple short-interval fires on community and functional trait-based regeneration in boreal Alaska.* Remote. April 2021.
 - Front Range Student Ecology Symposium. Hayes K, Buma B. *Repeat short-interval fires drive changes in forest structure, composition and carbon in Interior Alaska.* Remote. March 2021. **[Presentation Award]**
- 2020** International Association of Landscape Ecology: North America. Hayes K, Buma B. *Interacting effects of herbivory and short-interval reburns on successional trajectories in boreal Interior Alaska.* Remote. May 2020.
- 2019** Association for Fire Ecology. Hayes K, Buma B. *Continued short-interval fires overwhelm serotinous resilience regardless of topographic variation.* Tucson, AZ. November 2019.
- UC Denver Integrative Biology Graduate Seminar Series. Hayes K. *Repeat short-interval fires in boreal cause continued ecosystem change.* Denver, CO. October 2019.
 - NEON Science Summit. Hayes K. *Using NEON data to identify ecological tipping points across spatial/temporal scales.* Boulder, CO. October 2019.
 - International Association of Landscape Ecology: North America. Hayes K, Buma B. *Landscape Context mediates the effect of shortening fire intervals in boreal systems.* Fort Collins, CO. April 2019.
- 2017** Association of Pacific Coast Geographers. Hayes K, Gavin D. *Reconstructing paleofire in old growth coast redwood forests in Northern California using pyrogenic charcoal and soil carbon.* Chico, CA. October 2017. **[Presentation Award]**
- Ecological Society of America. Hayes K, Gavin D. *Reconstructing a fire history in coast redwood (*Sequoia sempervirens*) forests of Northern California.* Portland, OR. August 2017.
 - University of Oregon Graduate Research Forum. Hayes K. *Marine fog, climate change and coast redwood forests: Implications for management and research.* Eugene, OR. April 2017.
 - UO Climate Change Research Symposium. Hayes K, Gavin D. *Marine fog, climate change and coast redwood forests: Past, present and future.* Eugene, OR. April 2017.
- 2016** Midwest Undergraduate Geography Symposium. Hayes K, Williams J. *Fire history of Bonnett Lake, Ohio since the Last Glacial Maximum.* Minneapolis, MN. April 2016.
- University of Wisconsin Undergraduate Symposium. Hayes K, Williams J. *Compiling a continuous charcoal record of Bonnett Lake, Ohio since the Last Glacial Maximum.* Madison, WI. April 2016.

Conference Posters

*indicates student authors

- Association of Fire Ecology. Hayes K, Lauren L, Buma B. *Reburning impacts boreal forest regeneration trajectories across spatial scales.* New Orleans, LA. December 2025.
- American Geophysical Union. Hayes K, Honkaniemi J, Albrich K, Johnstone J, Hansen W. *Herbivore impacts on postfire boreal forest trajectories may not scale to impact landscape carbon balance.* Washington, DC. December 2024.
- Ecological Society of America. Hayes K, Hansen W, Honkaniemi J, Albrich K, Johnstone J. *Down the rabbit hole: Modeling the impacts of moose and hare herbivory on postfire carbon and community trajectories in Interior Alaska.* Long Beach, CA. August 2024.

- Forest Disturbance and Ecosystem Dynamics Symposium. Hayes K, Hoffman C, Lucash M, Buma B. *Short-interval reburning changes fuel structure, carbon storage and fire behavior of boreal forests*. Berchtesgaden National Park, Germany. September 2022. **[Poster Award]**
- International Fire Ecology and Management Congress. Hayes K, Hoffman C, Ziegler J, Buma B. *Continued short-interval reburning changes fuel structures of boreal forests*. Remote. December 2021.
- North American Carbon Program Open Science Meeting. Hayes K, Buma B. *Recovery of aboveground biomass and soil carbon after multiple short-interval disturbances in boreal Interior Alaska*. Remote. March 2021.
- American Geophysical Union. Hayes K, Buma B. *Effects of spatial heterogeneity on successional trajectories following repeat disturbances in Boreal Interior Alaska*. San Francisco, CA. Dec 2019.
- American Geophysical Union. Olson K*, Buma B, Hayes K. *Fine-scale observations of permafrost after repeat fires in Interior Alaska*. San Francisco, CA. December 2019.
- American Geophysical Union. Kodicherla V*, Shabaga J, Vogel J, Buma B, Hayes K. *Soil respiration in very high frequency Boreal wildfires as a function of species*. San Francisco, CA. December 2019.
- University of Colorado Denver Research & Creative Activities Symposium. Hayes K, Buma B. *The implications of increasing fire frequency in boreal forests in Interior Alaska*. Denver, CO. April 2019.
- University of Oregon Graduate Research Forum. Hayes K, Gavin D. *Fire and carbon cycling in old growth Coast redwood*. Eugene, OR. May 2018.
- Oregon Joint Campus Conference. Hayes K, Hendricks L, Gavin D. *Forests with naturally infrequent fire: their resilience and susceptibility to impacts by people & climate change*. Eugene, OR. May 2017.
- American Association of Geography. Hayes K, Gavin D. *Marine fog and climate change in Coast redwood (Sequoia sempervirens) forests: Implications for management & research*. Boston, MA. April 2017.

FUNDING

- 2026** Department of Defense Strategic Environmental Research and Development Fund (SERDP)
Boreal Burning Baselines: Historical variability and fire-related thresholds to ecosystem transformation
 Higuera P, Chipman M, Nanavati W, Hoffman C, **Hayes K**
 \$78,344 of \$1.6m
- 2025** Climate Science and Communication Fund Program
Surge Climate Talent Fund
 \$100,000
- Alaska Climate Adaptation Science Center Ambassador
Two fires, one stone: Identifying co-benefits of optimizing Alaska fire management for climate adaptation
 \$20,000
- 2024** Joint Fire Science Program Collaborative Proposal*
Informing effective strategies for managing carbon with increasing fire in boreal forests of Interior Alaska
 Hansen W, **Hayes K**, Hall J, Rogers B
 \$499,463
 * Cut by DOGE in Spring of 2025
- 2022** NSF Collaborative Proposal
The Past, Present and Future of Boreal Fire Feedbacks
 Buma B, Higuera C, Hoffman C, Chapman M, **Hayes K** [PhD Co-author]
 \$276,934 of \$1.4m
- NSF Postdoctoral Research Fellowship
Investigating fire-biotic disturbance interactions and their sensitivity to climate in the North American boreal biome

Hayes K [PI], Hansen W, Johnstone J
\$321,333

2021 UC Denver Integrative and Systems Biology Merit Grant
\$15,000

Colorado STEM Graduate Grant
\$1,250

2019 Joint Fire Science Program Graduate Research Innovation Award
Evaluating Flammability across Reburns in Interior Alaska
Hayes K, Buma B
\$25,000

Submitted / in review

Dept. of Defense Strategic Environmental Research and Development Program (SERDP)
Trialing RAD (Resist, Accept, Direct) Fire Management Options for Wildfire-caused transformation in Boreal Interior Alaska

Littell J, Rupp S, **Hayes K**

NSF FIRE-NET
Building a translational Alaska Fire Science Community of Practice
Hayes K, Schmidt J, Trainor S

TEACHING AND MENTORING EXPERIENCE

Graduate Courses

- Ecosystem Ecology, Primary Instructor, University of Alaska Fairbanks (*Spring 2026*)

Undergraduate Courses

- Ecosystem Ecology, Primary Instructor, University of Alaska Fairbanks (*Spring 2026*)
- Principles of Ecology, Instructor of Record, University of Colorado Denver (*Summer 2021*)
- General Biology: Organisms to Ecosystems, Teaching Assistant, University of Colorado Denver (*Spring 2022, Fall 2022*)
- Biostatistics, Teaching Assistant, University of Colorado Denver (*Fall 2019*)
- The Professional Geographer, Teaching Assistant, University of Oregon (*Spring 2018*)
- Quantitative Data Analysis, Teaching Assistant, University of Oregon (*Winter 2017*)
- Climatology, Teaching Assistant, University of Oregon (*Spring 2017*)
- Global Environmental Change, Teaching Assistant, University of Oregon (*Winter 2016*)
- Biogeography, Teaching Assistant, University of Oregon (*2 semesters, Fall 2016 & 2017*)

Guest Lecturer

- **Introduction to Fire Ecology**, Disturbance Ecology, UC-Denver (2020)
- **Introduction to Landscape Ecology**, Principles of Ecology, UC-Denver (2019)

Student Supervision

Field and lab assistants (12)

NSF Research Experience for Undergraduates (2)

2019 – 2020 Vishnusai Kodicherla - Tree impacts on soil respiration
Kristin Kraus – Relationship between reburn frequency & active layer depth
Outcomes: Two presentations, one publication in revision

Student committees (1)

2021 – 2022 Elizabeth Kuhn (B.S., University of Colorado Denver)
Community-scale indoor vertical hydroponic farms as a solution to food insecurity in urban settings

Curriculum Development

- Contributed data and feedback to updates of the *Wildfire Explorer (WFX) environment*, a teacher focused wildfire spread simulator build by the Concord Consortium (2023 – 2024)
- Co-developed *The Professional Geographer*, a professional development course in the Department of Geography at the University of Oregon (2017 – 2018)
- Contributed data and participated in educational [videos](#) of middle and high school lesson plan on boreal carbon cycling. Produced by ScienceLive Educational Modules (2019)

Pedagogy Training

- AGU Postdoctoral Research Fellow. AGU Leadership Academy and Network for Diversity and Inclusion in the Geosciences (LANDInG) (2023 – 2025)
- Pedagogy. Course at UC Denver, Department of Biology (2019 – 2020)
- Graduate Teaching Training. University of Oregon (2016)

AWARDS

2023 Ecological Society of America Paleoecology Travel Award. **\$300.**

2022 UC Denver Dept. of Biology Doctoral Teaching Award. **\$2,500.**
U.S. Carbon Program Leadership Award. **\$1,000.**
International Association of Landscape Ecology Presentation Award. **\$300.**
UC Denver Dept. of Biology Graduate Excellence in Research Award. **\$250.**

2021 International Boreal Forest Research Association Presentation Award. **\$400.**
Front Range Student Ecology Symposium Presentation Award. **\$150.**
Ecological Society of America Graduate Student Policy Award

2020 Polanki Graduate Achievement Award. **\$1,200.**

2019 - 2022 UC Denver Department of Integrative Biology Travel Award(s). **\$1,900.**
UC Denver College of Liberal Arts and Sciences Travel Award(s). **\$1,000.**
UC Denver Graduate School Travel Award(s). **\$500.**
Association of Fire Ecology Graduate Travel Award(s). **\$595.**

2018 Robert J. Leonard Memorial Award. **\$1,200.**

2017 Christopherson Award for Excellence in Applied Geography/Earth Systems Science. **\$300.**
UO Graduate Research Forum Oral Session Award. **\$75.**
Association of Pacific Coast Geographer's Travel Award. **\$200.**

SERVICE AND ENGAGEMENT

Stakeholder Engagement

- Alaska Interagency Fuels Workshop. Hosted by Alaska Fire Science Consortium and Alaska Wildland Fire Coordinating Group. (March 2025)
- Alaska National Wildlife Refuges Carbon Workshop. Hosted by US Fish & Wildlife and Northern Latitudes Partnerships. (March 2024)
- Alaska Wildfire Roundtable. Hosted by Alaska Venture Fund and Environmental Defense Fund. (October 2023)
- Alaska Fall/Spring Interagency Fire Operations Meeting(s). Hosted by Alaska Fire Science Consortium and Alaska Wildland Fire Coordinating Group. (2019 – Present)
- Bonanza Creek Community Symposium(s). Hosted by Bonanza Creek Long Term Ecological Research Program (LTER) (2019 – Present)

Collaborative Work

- Member. **Material Legacy Effects Synthesis Working Group**. LTER Network. (2025 – Present)
- Member. **Fire, Consumer, and Modeling Working Groups**. Bonanza Creek LTER (2023 – Present)
- Member. **Aspen Wildfire Pathogen Working Group**. Institute of Arctic Biology, University of Alaska Fairbanks (2023 – Present)
- Visiting Scientist. **Natural Resources Institute Finland (LUKE)**. (Spring 2023)

Service Activities

- Postdoc Representative. **Strategic Planning Committee**, Cary Institute of Ecosystem Studies (2023)
- **Postdoctoral Hiring Committee**, Bonanza Creek LTER (2022 – 2023)
- Graduate Representative. **Office of the Dean Graduate Advisory Group**, University of Colorado Denver (2020 – 2021)
- Student Representative. **International Association of Landscape Ecology North America** (2019 – 2021) **[Elected]**
- Graduate Representative. **Department of Biology Graduate Advisory Committee**, University of Colorado Denver (2019 – 2020)
- Department Representative. **Graduate Teaching Fellows Federation**, University of Oregon. (2016 – 2018)

Special sessions and Workshops

- Session Organizer - *Resilient natural systems in the paleo-record*. AGU, Washington DC (2025)
- Participant - *LTER Science Council*. Fairbanks, AK (2024)
- Participant - *NEON-NCAR Community Workshop*. Boulder, CO (2023)
- Panelist - *Advice from the Field: Practical Skills, Coping with Accessibility Challenges, and What Else You Need to Know*. ESA, Montreal, CA (2022)
- Session Organizer - *Exploring the context and implications of departures from historic fire frequency across ecosystems*. IALE-NA, Remote (2021)
- Session Organizer - *Water in the Pacific Midwest: Past and Future*. Graduate Student Research Forum, University of Oregon (2017) **[Organized Session Award]**

Volunteering

- Judge, *Hudson Data Jam Competition*, Cary Institute of Ecosystem Studies (2023)
- Judge, *Outstanding Student Presentation Award*, AGU Biogeosciences (2023)
- Visiting Guest Scientist, *Meet a Scientist Night*, Sitka Sound Science Center (2019)
- Abstract Reviewer, *Undergraduate Research Forum*, University of Oregon (2018)
- Moderator, *Oregon State Geography Bee*, National Geographic Society (2018)

Policy and Engagement Training

- Communicating with Journalists and the Media. COMPASS Training (2024)
- Strategic Science Communication. COMPASS Training (2024)

- Graduate Student Policy Award. Ecological Society of America (2021)
- Colorado Science Policy Summit. University of Colorado (2019)

Open Science and Quantitative Training

- Reproducible Practices for Arctic Research. NSF Arctic Data Center Workshop (2023)
- Foundations of Open Science Skills. CyVerse Workshop (2020)
- Using Google Earth Engine for Landscape Ecology. IALE North America Workshop (2020)
- Assimilating Long-Term Data into Ecosystems Models. PAIEON Workshop (2016)

Field Training

- Pest and Pathogen Workshop. Bonanza Creek LTER (2024)
- Wilderness First Aid License. (2022)
- Field Safety Training. National Science Foundation Polar Office (2018)

Journal and Grant Reviews

- *Nature, Plant and Soil, Global Change Biology, Ecosphere, OneEarth, Arctic, Antarctic, and Alpine Research, Ecosystems, Combustion Science and Technology, Forest Ecology and Management, Geoscientific Model Development, American Journal of Botany, International Journal of Wildland Fire, Conservation Biology, Ecology Letters*

Media

- **Hayes, K.** 2025. [A love letter to the Milwaukee Public Museum](#). Shepherd Express (February 27, 2025).